

Statistical and Computational Models

Lecture 10: Count outcomes and comparing models

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Roadmap

Counts

Overdispersion

Comparing nested models

Comparing non-nested models

Counting protests

Very often we encounter outcomes that count events:

$$Y_i \in \{0, 1, 2, \dots\}$$

Examples in political science:

- Bills passed in a session
- Violent events in a civil war
- Protests in a year

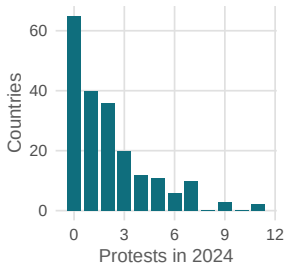
Why not just fit OLS to Y_i ? Three downsides:

1. Fitted counts can be **negative**.
2. A constant β_j is implausible: near zero, counts **cannot fall** far.
3. Counts' variance **grows with the mean**: heteroskedastic errors.

Counting protests

Suppose our running example also recorded the number of protests in 2024:

protests	regime	e_autocracy	lgdp	oil
2	Democracy	0	9.23	0
2	Democracy	0	8.32	0
0	Closed autocracy	0	6.49	1
1	Electoral autocracy	1	7.56	1



Bars: our 205 countries. Many zeros, and a long right tail.

We want to model `protests` with the same covariates as before, `oil` and `lgdp`, plus regime type, as `e_autocracy`: 1 for an electoral autocracy, 0 for a closed autocracy or a democracy.

The Poisson distribution

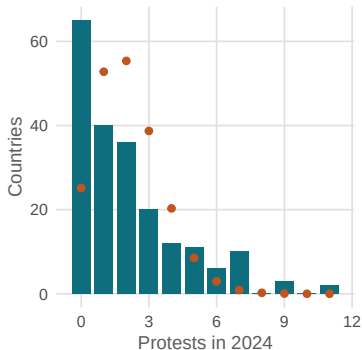
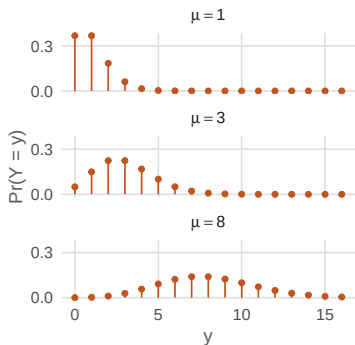
Suppose events occur 1. continuously, 2. independently, and 3. with constant probability. Then the count in a fixed period is **Poisson**:

$$Y_i \sim \text{Poisson}(\mu), \quad \mathbb{P}(Y_i = y) = \frac{\mu^y}{y!} e^{-\mu} = \frac{\mu}{1} \cdot \frac{\mu}{2} \cdots \frac{\mu}{y} e^{-\mu}$$

Read: the k th event multiplies by μ/k , so counts **beyond** μ grow rarer.

Note: one parameter, μ , is both the mean and the variance.

The Poisson distribution



Left: the mean moves the peak, and the spread **grows with it**.

Right: our 205 countries (bars), and a Poisson with the same mean, $\bar{y} = 2.10$ (points).

Poisson regression

Recall: every exponential-family distribution has a **canonical parameter**.
For the Poisson, it is $\log \mu$:

Y_i	Mean	Canonical parameter	Canonical $\mathbf{g}$	Model
Normal	μ	μ	identity	OLS
Bernoulli	π	$\log\{\pi/(1 - \pi)\}$	logit	logit
Poisson	μ	$\log \mu$	log	Poisson

So $Y_i \sim \text{Poisson}(\mu_i)$ with $\log \mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}$:

$$\mathbb{E}[Y_i \mid \mathbf{x}_i] = \text{Var}[Y_i \mid \mathbf{x}_i] = \mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta})$$

Read: the expected count comes from the exponential of a weighted sum of the covariates $\mathbf{x}_i$.

Aside: logit vs. Poisson

Recall that a GLM has two pieces: a **distribution** for the outcomes, and a **link** that connects the linear model $\mathbf{x}_i^\top \boldsymbol{\beta}$ to the distribution's parameter, keeping it in range.

Model	Distribution	Outcomes	Quantity	Range	Inverse link
Logit	Bernoulli	$\{0, 1\}$	$\mathbb{P}(Y_i = 1)$	$(0, 1)$	$\text{logit}^{-1}(\mathbf{x}_i^\top \boldsymbol{\beta})$
Ordered logit	Categorical*	$\{1, \dots, J\}$	$\mathbb{P}(Y_i \leq j)$	$(0, 1)$	$\text{logit}^{-1}(\zeta_j - \mathbf{x}_i^\top \boldsymbol{\beta})$
Multinomial logit	Categorical*	$\{1, \dots, J\}$	$\mathbb{P}(Y_i = j)$	$(0, 1)$	$e^{\mathbf{x}_i^\top \boldsymbol{\beta}_j} / \sum_k e^{\mathbf{x}_i^\top \boldsymbol{\beta}_k}$
Poisson	Poisson	$\{0, 1, 2, \dots\}$	$\mu_i = \mathbb{E}[Y_i]$	$(0, \infty)$	$\exp(\mathbf{x}_i^\top \boldsymbol{\beta})$

Read: the distribution says how likely each **outcome** is (and so fixes the variance); the inverse link maps $\mathbf{x}_i^\top \boldsymbol{\beta}$ into the **right range**.

So Poisson tackles a slightly different problem to the logits: they needed a floor and a ceiling; a count needs only a **floor**.

*Categorical: a Bernoulli with J outcomes, $\mathbb{P}(Y_i = j) = \pi_{ij}$, where $\sum_{j=1}^J \pi_{ij} = 1$.

The Poisson likelihood

Assume that Y_i is drawn from a Poisson, with μ_i depending on covariates $\mathbf{x}_i$ through parameters β (that we care about).

Write down the log-likelihood given your assumption:

$$L(\beta; \mathbf{Y}, \mathbf{X}) = \prod_{i=1}^n \frac{\mu_i^{y_i}}{y_i!} e^{-\mu_i}, \quad \mu_i = \exp(\mathbf{x}_i^\top \beta)$$
$$\ell(\beta) = \sum_{i=1}^n \left\{ \underbrace{y_i \mathbf{x}_i^\top \beta}_{y_i \log \mu_i} - \underbrace{\exp(\mathbf{x}_i^\top \beta)}_{\text{expected count}} - \underbrace{\log y_i!}_{\text{no } \beta} \right\}$$

Read: each country's $y_i \log \mu_i - \mu_i$ peaks at $\mu_i = y_i$; $\log y_i!$ does not move with β .

Fitting Poisson regression in R

Using `glm()`:

```
fit_pois <- glm(protests ~ lgdp + e_autocracy + oil, data = countries,  
               family = poisson)  
round(coef(summary(fit_pois))[, 1:2], 2)
```

	Estimate	Std. Error
(Intercept)	-2.10	0.46
lgdp	0.30	0.05
e_autocracy	0.63	0.11
oil	-0.03	0.11

Note: `family = poisson` imposes the log link by default.

The rows are the **coefficients** $\hat{\beta}$, which, given the log link, linearly map $\mathbf{x}_i$ to the **log** of the expected count.

What is β in Poisson regression?

Consider a single β_m inside our vector β :

$$\beta_m = \underbrace{\frac{\partial}{\partial x_{im}}}_{\text{change in } x_{im}} \log \underbrace{\mathbb{E}[Y_i | \mathbf{x}_i]}_{\text{expected count}}$$

Read: the change in the log **expected count** per unit of x_{im} .

Adding β_m on the log scale **multiplies** the expected count by e^{β_m} . Raise x_{im} by one, other covariates fixed:

$$\frac{\mathbb{E}[Y_i | x_{im} + 1]}{\mathbb{E}[Y_i | x_{im}]} = \frac{\exp(\dots + \beta_m x_{im} + \beta_m)}{\exp(\dots + \beta_m x_{im})} = e^{\beta_m}$$

So e^{β_m} is a **ratio of expected counts**, at one unit more of x_{im} over at x_{im} , for every country (aka the incidence rate ratio).

Interpreting $\hat{\beta}_m$ in Poisson regression

Some guidance for interpreting a single $\hat{\beta}_m$:

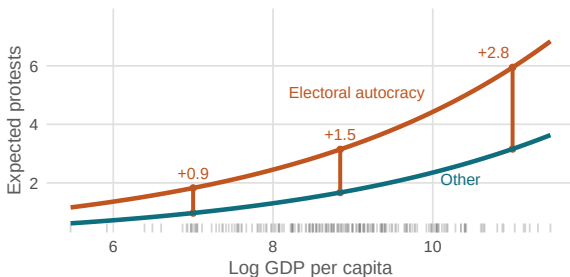
1. Read the **sign**: `e_autocracy` raises the expected count.
2. Exponentiate: $e^{\hat{\beta}_{\text{e_autocracy}}} = 1.88$. Electoral autocracies expect 1.88 times as many protests, at the same `lgdp` and `oil`.
3. For a change of size `c`, $e^{c\hat{\beta}_m}$.

Note: steps 2 and 3 work because of the **log link**: on the log scale, a change in x_{im} adds the same $\hat{\beta}_m$ for every country.

Interpreting regime type, *ceteris paribus*

As with the logit, we want **quantities of interest** in the outcome's units: there probabilities, here counts.

Each curve is a fitted value, $\hat{\mu} = \exp(\mathbf{x}^\top \hat{\beta})$: the **expected count** at that `lgdp`, with `oil = 0` and `e_autocracy` off or on:



The ratio of the two curves is 1.88 at every `lgdp`. The **gap** in protests is not: it grows with the expected count.

Interpreting regime type, *ceteris paribus*

Averaging the gap over the observed countries gives one **APE**:

```
marginaleffects::avg_comparisons(fit_pois, variables = "e_autocracy")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
1.54	0.313	4.91	<0.001	20.1	0.925	2.15

Term: e_autocracy

Type: response

Comparison: 1 - 0

Read: electoral autocracies expect about 1.5 more protests a year than otherwise similar countries, on average over our countries.

Again, the APE has an advantage over raw or exponentiated coefficients: it is a **quantity of interest** in protests, the units readers care about.

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Overdispersion

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The variance is usually wrong

The Poisson assumes $\mathbb{E}[Y_i | \mathbf{x}_i] = \text{Var}[Y_i | \mathbf{x}_i]$. But for many counts, $\mathbb{E}[Y_i | \mathbf{x}_i] < \text{Var}[Y_i | \mathbf{x}_i]$. This is called **overdispersion**.

Why might protests be overdispersed?

- **Unobserved heterogeneity**: two electoral autocracies with the same `lgdp` and `oil` can still differ, e.g. in how organised the opposition is. Each has its own rate, and the rates average to μ_i .
- **Contagion**: one protest makes the next more likely, so protests come in bursts.

Read: neither process necessarily moves the **mean**, but both increase the spread of counts, breaking the equality between mean and variance.

The variance is usually wrong

Two questions: is $\hat{\beta}$ still right? Are its standard errors?

The **estimate**: $\hat{\beta}$ sets the slope of ℓ to zero,

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n (y_i - \mu_i) \mathbf{x}_i = \mathbf{0}$$

Read: at the true β , each $Y_i - \mu_i$ averages to zero whenever $\exp(\mathbf{x}_i^\top \beta)$ is the right mean, whatever the variance. So $\hat{\beta}$ is still **consistent**.

The variance is usually wrong

The **standard errors**: $\hat{\beta}$ is where the slope crosses zero, so the more the slope moves from sample to sample, the more $\hat{\beta}$ moves.

The slope contains Y_i , so its spread involves $\text{Var}[Y_i | \mathbf{x}_i]$. The curvature does not:

$$\text{Var}\left[\frac{\partial \ell}{\partial \beta}\right] = \sum_{i=1}^n \text{Var}[Y_i | \mathbf{x}_i] \mathbf{x}_i \mathbf{x}_i^\top \quad \text{vs.} \quad -H(\beta) = \sum_{i=1}^n \mu_i \mathbf{x}_i \mathbf{x}_i^\top$$

The Poisson's standard errors use the curvature on the right in place of the spread on the left. These match when $\text{Var}[Y_i | \mathbf{x}_i] = \mu_i$ for every i .

With overdispersion the left is bigger: the standard errors are **too small**.

The variance is usually wrong

There are two fixes we can apply to the Poisson regression itself:

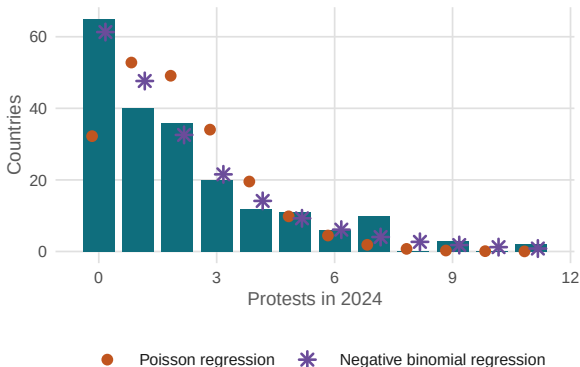
1. **Quasi-Poisson** keeps the mean but lets the variance be $\phi\mu_i$, and estimates ϕ from the data ($\hat{\phi} = 2.39$).
2. The **sandwich**, $\text{Var}[\hat{\beta}] \approx (-H)^{-1} \widehat{\text{Var}}\left[\frac{\partial \ell}{\partial \beta}\right] (-H)^{-1}$, estimates the middle from the residuals $(y_i - \hat{\mu}_i)^2$.

	Estimate	Poisson se	Quasi-Poisson se	Sandwich se
(Intercept)	-2.098	0.458	0.708	0.674
lgdp	0.295	0.048	0.075	0.070
e_autocracy	0.632	0.110	0.170	0.173
oil	-0.035	0.110	0.171	0.166

Both give us the same $\hat{\beta}$, but more conservative standard errors.

Negative binomial

Rather than rescuing the Poisson regression, let's try a different **probability distribution**: one with a likelihood whose variance can grow faster than its mean.



Negative binomial

The **negative binomial** (NB2) adds one parameter, θ :

$$\mathbb{E}[Y_i | \mathbf{x}_i] = \mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}), \quad \text{Var}[Y_i | \mathbf{x}_i] = \mu_i + \mu_i^2 / \theta$$

Read: the mean is the **same** as the Poisson's. The μ_i^2 term lets the variance **outrun** the mean – more zeros and a longer right tail than a Poisson with that mean. As $\theta \rightarrow \infty$, we get the Poisson back.

MASS: :glm.nb(): $\hat{\theta} = 1.42$.

Negative binomial

With θ fixed, the negative binomial is an exponential family too:

Y_i	Mean	Canonical parameter	Link in $\mathbf{R}$	Model
Normal	μ	μ	identity	OLS
Bernoulli	π	$\log\{\pi/(1 - \pi)\}$	logit	logit
Poisson	μ	$\log \mu$	log	Poisson
NB (θ fixed)	μ	$\log\{\mu/(\mu + \theta)\}$	log	NB

`glm.nb()` uses the **log** link, not the canonical one. So the same three steps for $\hat{\beta}_m$ as in Poisson regression.

Fitting GLMs in R

Family or function	Link	Variance	Model
<code>gaussian</code>	identity	σ^2	normal linear
<code>binomial</code>	logit (or probit)	$\mu(1 - \mu)$	logit, probit
<code>poisson</code>	log	μ	Poisson
<code>quasipoisson</code>	log	$\phi\mu$	overdispersed Poisson
<code>MASS::glm.nb</code>	log	$\mu + \mu^2/\theta$	negative binomial
<code>MASS::polr</code>	cumulative logit	–	ordered logit
<code>nnet::multinom</code>	baseline-category logit	–	multinomial logit

`{marginaleffects}` and `{modelsummary}` take every fit in the table.

But both can misbehave when robustifying standard errors (e.g. `vcov = "HC3"` fails for `polr` and `multinom`, and `modelsummary()` silently ignores `vcov` for `multinom`). Check the documentation, and check that the standard errors actually changed.

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Comparing two models

Suppose we want to know whether `e_autocracy` belongs in the model. One approach is to fit the negative binomial with and without it:

$$\ell(\text{with}) = -388.2, \quad \ell(\text{without}) = -394.6$$

The **likelihood-ratio** statistic compares the two:

$$\text{LR} = -2 \log \frac{L(\text{without})}{L(\text{with})} = 2 [\ell(\text{with}) - \ell(\text{without})] = 12.8$$

Read: how much the **likelihood** rises when we add `e_autocracy`.

Adding a parameter can never lower it, so the question is whether the rise beats chance.

If the dropped parameter is truly zero, $\text{LR} \overset{a}{\sim} \chi_1^2$ (one parameter dropped). Here $p < 0.001$. In R, `anova(fit_nb0, fit_nb)`.

Comparing two models

The LR test answers **one** question: does adding `e_autocracy` raise the in-sample likelihood by more than chance?

It does not answer:

- How **big** is the effect?
- Does the model **predict** new countries better?
- Is the effect **causal**?

So the LR test applies only under two conditions: the models are **nested** (one is a special case of the other), and we want to know which **fits the data better**. That is not the same as knowing which is the better model!

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Comparing non-nested models

The likelihood-ratio test needs **nested** models. Suppose we fit both a negative binomial and OLS to `protests`.

The **Akaike Information Criterion**:

$$\text{AIC} = -2\hat{\ell} + 2d$$

Read: the maximised log-likelihood $\hat{\ell}$, charged 2 for each of the d parameters. Smaller is better.

The **Bayesian Information Criterion**, $\text{BIC} = -2\hat{\ell} + d \log n$, charges $\log n$ per parameter instead: a heavier penalty once $n > 7$ (here $\log 205 = 5.3$).

Note: AIC and BIC are **in-sample** adjustments to the training loss – cheap stand-ins for cross-validation.

Comparing non-nested models

Both models fit to **protests** (OLS scored as a normal rounded to each count, so the likelihoods are comparable):

Model	Parameters	ℓ	AIC	BIC
Negative binomial	5	-388.2	786.5	803.1
OLS (normal errors)	5	-454.2	918.4	935.0

Read: the negative binomial wins on both, by a wide margin.

Why? OLS puts, on average, 13% of its probability on **negative** protest counts.

Cross-validation for counts

Let's try cross-validation instead: the negative binomial against OLS, scoring the held-out fold $\mathcal{I}_j$ two ways:

$$\text{MSE} = \frac{1}{n_j} \sum_{i \in \mathcal{I}_j} (y_i - \hat{\mu}_i)^2, \quad -2\ell = \frac{1}{n_j} \sum_{i \in \mathcal{I}_j} -2 \log \hat{p}(y_i)$$

Read: MSE scores the predicted **mean**; -2ℓ scores the predicted **distribution**, $\hat{p}$.

```
for (j in 1:K) {  
  tr <- countries[fold != j, ]  
  va <- countries[fold == j, ]  
  fit_cv <- MASS::glm.nb(protests ~ lgdp + e_autocracy + oil, data = tr)  
  mu_cv <- predict(fit_cv, va, type = "response")  
  mse[j] <- mean((va$protests - mu_cv)^2)  
  ll[j] <- -2 * mean(dnbinom(va$protests, size = fit_cv$theta,  
                             mu = mu_cv, log = TRUE))  
}
```

Cross-validation for counts

With $K = 10$ folds:

Model	CV MSE	CV -2ℓ	AIC / n
Negative binomial	5.12	3.84	3.84
OLS (normal errors)	5.14	4.52	4.48

Which model wins under MSE? Neither: they predict nearly the **same mean**.

Under -2ℓ , the negative binomial: it scores the **whole distribution**, and OLS wastes probability on impossible counts. AIC/ n agrees.

References

- Rodríguez, G. *Generalized Linear Models*, Princeton lecture notes – **ch. 4**, Addendum 4.A. grodriguez.github.io/glms.